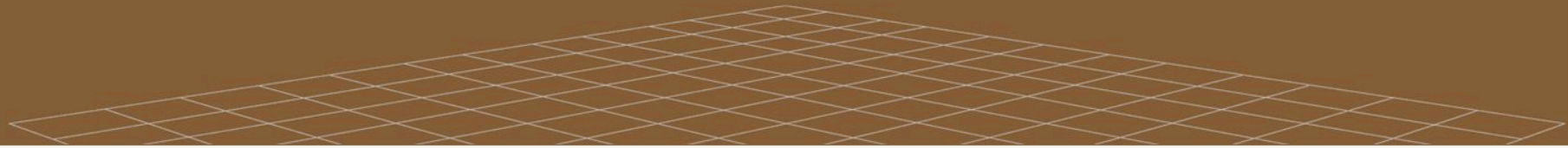
Growth and development Overview

Third Stage

Dr. Moustafa Alshammari

Growth and Developments



Adolescent Period Objectives:

Upon Completion and mastery of adolescent period of development the student will be able to:

- Discuss the phases of adolescent development.
- Explain the essential changes in physical development of adolescents
- Explain psychosocial development of adolescents, according to Erikson.

Cont. Adolescent Period Objectives

4. Explain the essential changes in cognitive development during adolescence according to Piaget.
5. Describe moral development of adolescents
6. Discuss spiritual development of adolescents.
7. Identify common health hazards and concerns of adolescents.
8. Discuss nursing implications related to common health concerns of adolescents.

Adolescent Period

- Transitional period from childhood to adulthood.
- Adolescents often feel a sense of stress and anxiety throughout this period.
- Nurses must understand the types of physical and psychological changes that face the adolescent and the sources of frustration they meet in society.

- During this period , the person becomes physically and psychologically mature and acquires a personal identity.
- At the end of this critical period in development, the person is ready to enter adulthood and assume its responsibilities.
- The length of adolescence is culturally determined to some extent. North America adolescence is longer than in some cultures, extending to 18 to 20 years of age.

Puberty سن البلوغ

It is the first stage of adolescence in which sexual organs begin to grow and mature.

For girls 10 and 14 years.

For boys 12 and 16 years.

Phases of adolescent:

- Early Adolescence: Age from 12 to 13 years.
- Middle Adolescence: Age 14 to 17 years.
- Late Adolescence: Age 17 to 20 years.

- N.B: Late adolescence is a more stable stage than the other two.
- In this phase, adolescents are involved mostly with planning their future and economic independence.

Physical Development:

- During puberty, growth is markedly accelerated compared to the slow, steady growth of the child.
- This period is marked by sudden and dramatic physical changes.
- This sudden growth is referred to as the adolescent growth spurt. طفرة
- Because the growth spurt begins earlier in girls, many girls surpass boys in height at this time.

Physical Growth

- It is continues throughout adolescence.
- For boys *Growth is fastest at about 14 years, and the maximum height is often reached at about 18 or 19 years. Some men add another 1 or 2 cm to their height during their 20s.
- For girls *Growth is fastest at about age 12, they reach their maximum height at about 15 to 16 years.
- Factors that influence *Physical growth during adolescence are numerous , such as heredity, nutrition, medical care, illness, physical and emotional environment, family size, and culture.

- **Growth is noted first in the musculoskeletal system.**

This growth follows a sequential pattern:

- **The head, hands, and feet are the first to grow to adult status.**
- **Next, the extremities reach their adult size.**
- **Because the extremities grow before the trunk, the adolescent looks leggy, awkward, and uncoordinated.**
- **After the trunk grows to full size, the shoulders, chest, and hips grow.**
- **Skull and facial bones also change proportions. The forehead become more prominent and the jawbones develop.**

Poor posture

Poor posture is a common problem during adolescence. The risk for postural problems increases among this age group **because weight gains may precede a corresponding strengthening of postural muscles.**

Glandular Changes

- **Eccrine glands** increase their secretions and become fully functional during puberty. **These glands, are** found over most of the body, they produce sweat.
- **Apocrine glands** develop in the axillae, anal and genital areas, external auditory canals and around the umbilicus and the areola of the breasts. **Apocrine sweat** is released onto the skin in response to emotional stimuli only.

- **Sebaceous glands** also become active under the influence of androgens in both males and females. **Sebaceous glands,** which secrete **sebum,** become most active on the face, neck, shoulder, upper back, chest, and genitals.

  When these glands become plugged and inflamed, the result is **acne,** a common condition in adolescence.

Acne



Sexual Characteristics

1. **Primary sexual characteristics** relate to the organs necessary for **reproduction**, **such as** the testes, penis, vagina, and uterus.

2. **Secondary sexual characteristics** differentiate the male from the female but do not relate directly to reproduction. **Examples** are pubic hair growth, breast development, and voice changes.

Male puberty

- **The first noticeable sign that puberty** has begun in **male,** is the appearance of pubic hair.
- The **milestone** of male puberty is considered to be the **first ejaculation**, which **commonly occurs** at **about 14 years** of age.
- Fertility follows several months later. **Sexual maturity is achieved by age 18.**

Females puberty

- **The first noticeable sign of puberty** in females is the appearance of the **breast buds,** although the appearance of hair along the labia may precede this. **This usually starts at 8-13 years.**
- **The milestone** of female puberty is **menarche,** which occurs about 2 years after the breast bud appears. At first menstrual periods are scanty

- **Ovulation** is usually established **1 to 2 years after menarche.**

- **Female** internal reproductive organs reach adult size about age 18 to 20.

Cognitive Development

- Cognitive abilities mature during adolescence. Between age of 11 and 15, the adolescent begins according to **Piaget's The Formal operations stage of cognitive development**.

- Adolescents are able to think abstractly as well as logically.

- They think about their future and the parts they will play in society as adults.

- They also think about other people's thinking often critically.
- Consider possibilities, hypotheses.
- Can use **deductive** (general to specific) and **inductive** (specific to general) reasoning.

- **Adolescents are highly imaginative and idealistic**

- **The adolescents become more informed about the world and environment.**

- **Adolescents use achieving a sense of new information to solve everyday problems and can communicate with adults on most subjects.**

- **The adolescent's capacity to absorb and use knowledge is great. Usually select their own areas for learning, they explore interests from which they may evolve a career plan.**

Psychosocial Development:

- According to Erickson theory, Identity/ Identity diffusion is correct during the adolescent stage.
- A sense of identity means a sense of being at one with oneself as one grows and develops. Identity development is closely linked with the mastery of skills.
- Identity as well as overcoming a sense of identity diffusion represents the polarity الاختلافof this developmental phase.
- Adolescents are usually concerned about their bodies, their appearances, and their physical abilities. Hair styling, skin care, and clothes become very important.

- **Young adolescents experience a close attachment to their parents while they reach for new and more satisfactory contacts in association outside the family.**

- **They learn to interact with more and more people at a given times and they learn in which way they should relate to them.**

- **In this process they become individuals and members of their community.**

- **Adolescents develop their relationship with significant adult in his life.**
- **Identity development moves from the reality of being to the reality of becoming.**
- **Although sexual identification begins at about 3 or 4 years of age, it is a significant part of adolescence.**
- **The adolescent male strive to achieve masculine sexual identity**
- **The adolescent female strives to achieve a feminine sexual identity.**

Peer groups during adolescence assume great importance.

The peer group has a number of functions:

1. It provides a sense of belonging, pride, social learning, and sexual roles.

2. Most peer groups have well defined, sex-specific modes of acceptable behavior.

3. In adolescence, the peer groups change with age. They start as single-sex groups, evolve to mixed groups, and finally narrow to couples who share activities.

Health Promotion and Wellness:

Health maintenance visits and immunization:

- **The adolescent should receive routine health assessments, appropriate laboratory screening, and periodic dental care.**
- **Immunization required for other age periods and were not received they should be given at this time.**
- **Periodic vision and hearing screening.**
- **Obtaining and providing accurate information about sexual issues.**

Nutrition:

- The adolescent's need for nutrients and calories increases should be recognized, particularly during growth spurt.
- In particular, the need for protein, calcium, vitamin D, iron, and B vitamins increases during adolescence.
- Importance of healthy snacks and appropriate patterns of food intake and exercise should be discussed.
- Explain factors that may lead to nutritional problems (e.g. obesity, anorexia nervosa, and bulimia)

Skin care:

- Adolescents need teaching and guidance to help them deal with the changing needs of their bodies.
- Secretions from newly active sweat glands react with bacteria on the skin, causing **a pungent odor** (strong smelling).
- Teenagers need to practice good hygiene to be sure that clothes smell fresh and clean.

Rest/Sleep :

- Most adolescents require 8 to 10 hours of sleep each night to prevent undue fatigue and susceptibility to infections.
- A change in sleep pattern is common in adolescence.

Menstruation:

Girls need to be taught about the menstrual cycle and necessary self-care responsibilities.

Sexuality:

- Adolescents want to know about sex but are often uneasy about discussing these concerns with their parents.

- Nurses, schools, and the family need to provide accurate information.

- Parents who have established open communication regarding sexual changes and reproduction during the school age period are more likely to be asked questions and have discussions with their teenagers regarding sexual issues.

Common health hazards

- **Physical Fitness**:
- Many boys and girls become less active during adolescence. Lack of exercise affects both mental and physical health.
- Even moderate physical activity, such as brisk walking, bicycling, swimming, or yard work, has health benefits if done regularly for at least 30 minutes at most, and preferably all days of the week.

Nutrition and Eating Disorders:

- **Obesity:**

  Too little physical activity and poor eating habits are causes of obesity within a person's control.

  It Is a common problem of the preadolescent period and continues to be a problem in the adolescent period.

- **Anorexia Nervosa:**

It is characterized by:

Prolonged inability or refusal to eat

Rapid weight loss, and emaciation in persons who continue to believe they are fat.

Use of laxatives and diuretics to remain thin.

Anorexics may also induce vomiting

- **Bulimia Nervosa:** is an uncontrollable compulsion to consume enormous amount of food and then expel it by self induced vomiting or by taking laxatives.

Use and Abuse of Drugs:

- **Substance abuse**

  It is repeated, harmful use of a substance, usually alcohol or other drugs.

  Abuse can lead to **substance dependence:** Addiction, which may be physiological or psychological, or both, and is likely to continue to a harmful substance.

- **Gateway Drugs:**

  (Alcohol, Marijuana, and Tobacco): The three drugs are sometimes called Gateway Drugs because their use often leads to more addictive substances, such as cocaine and heroin.

Adolescent Pregnancy:

- **Risks to infants:**

  Two major risks are associated with adolescent pregnancies:

1. Preterm birth
2. Low-birth weight infants, are associated

- **Factors that contribute to having a low-birth weight infants are:**

1. Low socioeconomic status
2. Minimal education
3. Inadequate prenatal care
4. Poor weight gain
5. Poor nutrition, smoking
6. Use of recreational drugs.
7. Adolescent parents are not prepared to anticipate or recognize early signs of illness or household dangers.

Risks to adolescent mothers:

Many adolescents engage in risk-taking behaviors to establish their independence. These include:

- Smoking
- Taking drugs
- Drinking alcohol
- Becoming sexually active with the consequences of pregnancy and sexually transmitted diseases (STD)

Sexually Transmitted Diseases:

Specific STDs that are greatly increasing among adolescents are

- **Gonorrhea**
- **Chlamydia**
- **Genital herpes infections**
- **Genital warts caused by the human**
- **Papilloma virus (HPV)**
- **HIV.**

Health Problems Affecting Body Image of the Adolescent

- Acne
- Obesity
- Dental or vision problems
- Trauma

Information Needed by Adolescents Regarding Their Changing Bodies

- **Girls:** Handling their early menstrual periods with minimal apprehension
- **Boys:** Bodily functions including erections and nocturnal emissions
- **Boys and girls:** Changes in the opposite sex hormonal changes and changes in feelings

Important Psychosocial and cognitive developmental achievements of the Adolescent include:

- From Freud's latency stage to the genital stage
- From Erikson's industry versus inferiority to identity versus role confusion
- From Piaget's concrete operational thinking to formal operational thought

Nursing Care of the Adolescent in a Health Care Facility

- Provide supportive nursing care sensitive to the adolescent's needs
- Encourage as much participation as possible
- Clearly and honestly explain all procedures and treatments
- Protect privacy by providing screening and covering
- Provide a telephone and recreational areas to keep up social contacts with peers

Thank you

رمز الصف

cs5jm5o